# Graphing Polynomial Functions

Name _____

Algebra 2 Lesson 3-1 (fast track) Date _____ Period _____

**Learning goal:** I can predict the end behavior of a polynomial function from its degree and leading coefficient, and use a table of values to find its turning points and sketch its graph.

**Language goal:** I can describe a polynomial graph using the words *degree*, *leading coefficient*, *end behavior*, *turning point*, and *relative maximum / relative minimum*.

**On the quiz:** Quiz 2: end behavior from the degree and the leading coefficient (Example 2). Quiz 3: all the turning points on a graph (Example 3). Quiz 4, bonus: where a graph is increasing, decreasing, positive, or negative (Example 3 and the Practice). Quiz 1, a true statement about a degree-5 polynomial, and Quiz 5, what a $y$-intercept means in context: full Savvas quiz only.

**On the Topic 3 test:** Question 6 asks you to sketch a cubic from a table and end behavior (Examples 2 and 3). Question 9 asks which volume grows faster on an interval (Example 3).

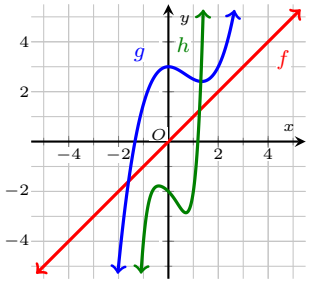
## Example 2: Understand End Behavior of Polynomial Functions

How do the sign of the leading coefficient and the degree of a polynomial affect the end behavior of the graph of a polynomial function?

Write the polynomial in **standard form** (terms from the greatest power of $x$ down). The **degree** is the greatest power; the **leading coefficient** is the coefficient of that term. The **end behavior** of a graph describes what happens to the function values as $x \to \pm\infty$.

**Odd degree**
**Positive leading coefficient**

$f(x) = x$
$g(x) = 0.5x^3 - x^2 + 3$
$h(x) = 2x^5 - x^2 - x - 2$

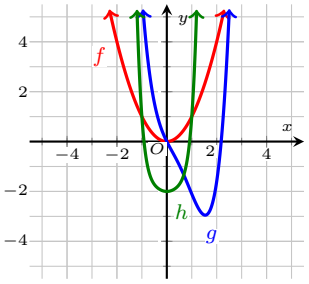



As $x \to \infty$, $y \to \infty$;
as $x \to -\infty$, $y \to -\infty$.

*like the parent* $f(x) = x$

**Even degree**
**Positive leading coefficient**

$f(x) = x^2$
$g(x) = 0.9x^4 - 2x^3 + x^2 - 2x$
$h(x) = 2x^6 + x^2 - 2$

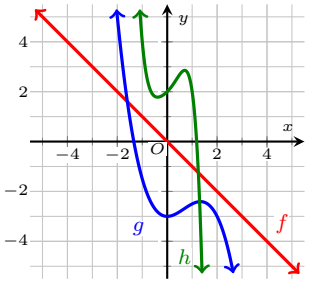



As $x \to \pm\infty$, $y \to \infty$.

*like the parent* $f(x) = x^2$

**Odd degree**
**Negative leading coefficient**

$f(x) = -x$
$g(x) = -0.5x^3 + x^2 - 3$
$h(x) = -2x^5 + x^2 + x + 2$

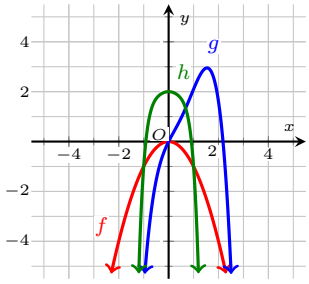



As $x \to \infty$, $y \to -\infty$;
as $x \to -\infty$, $y \to \infty$.

*like* $f(x) = -x$

**Even degree**
**Negative leading coefficient**

$f(x) = -x^2$
$g(x) = -0.9x^4 + 2x^3 - x^2 + 2x$
$h(x) = -2x^6 - x^2 + 2$




As $x \to \pm\infty$, $y \to -\infty$.

*like* $f(x) = -x^2$

**The rule.** The **degree** decides whether the two ends point the *same* way (even) or *opposite* ways (odd). The sign of the **leading coefficient** decides where the right end goes: + up, − down. The other terms do not change the ends: the leading term has the greatest exponent, so it has the greatest effect as $x \to \pm\infty$.

**Say it:** $g(x) = -0.5x^3 + x^2 - 3$ has **degree** _____, which is _____, and **leading coefficient** _____, which is _____. So its **end behavior** is: as $x \to \infty$, $y \to$ _____, and as $x \to -\infty$, $y \to$ _____.

## Try It 2: Use the leading coefficient and degree

Use the leading coefficient and degree of the polynomial function to determine the end behavior of each graph.

**a.** $f(x) = 2x^6 - 5x^5 + 6x^4 - x^3 + 4x^2 - x + 1$

degree _____ leading coefficient _____

As $x \to \infty$, $y \to$ _____

As $x \to -\infty$, $y \to$ _____

**b.** $g(x) = -5x^3 + 8x + 4$

degree _____ leading coefficient _____

As $x \to \infty$, $y \to$ _____

As $x \to -\infty$, $y \to$ _____

## Example 3: Graph a Polynomial Function

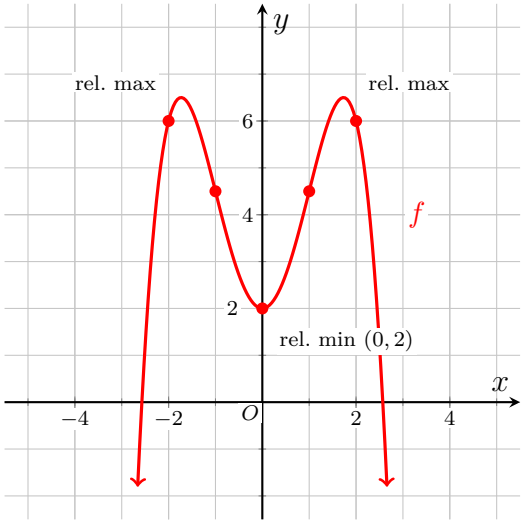
Consider the polynomial function $f(x) = -0.5x^4 + 3x^2 + 2$.

**A.** How can you use a table of values to identify key features and sketch a graph of the function?

1. Make a table of values. For example, $f(-2) = -0.5(16) + 3(4) + 2 = 6$.

| $x$ | $-3$ | $-2$ | $-1$ | $0$ | $1$ | $2$ | $3$ |
|---|---|---|---|---|---|---|---|
| $f(x)$ | $-11.5$ | $6$ | $4.5$ | $2$ | $4.5$ | $6$ | $-11.5$ |

2. Read the table left to right. The values of $f$ **increase** from $x = -3$ to $x = -2$, **decrease** from $x = -2$ to $x = 0$, increase from $x = 0$ to $x = 2$, and decrease from $x = 2$ to $x = 3$.
3. A **turning point** is where the values change from increasing to decreasing, or vice versa. The table shows turning points near $x = -2$, $x = 0$, and $x = 2$; it only estimates where they are.
4. End behavior: degree 4 (even), leading coefficient $-0.5$ (negative), so both ends approach $-\infty$.
5. Plot the points and sketch a smooth curve.




A point where the function has the least value over an interval is a **relative minimum**: near $(0, 2)$. A point where it has the greatest value over an interval is a **relative maximum**: near $(-2, 6)$ and $(2, 6)$. (A table only estimates: the smooth curve's relative maximums are at $\left(\pm\sqrt{3},\ 6.5\right)$, about $(\pm 1.73,\ 6.5)$.)

**B.** How can you use the graph to estimate the average rate of change over the interval $[-2, 0]$?

Average rate of change $= \dfrac{f(b) - f(a)}{b - a} = \dfrac{f(0) - f(-2)}{0 - (-2)} = \dfrac{2 - 6}{0 - (-2)} = \dfrac{-4}{2} = -2.$ Substitute $(-2, 6)$ and $(0, 2)$.

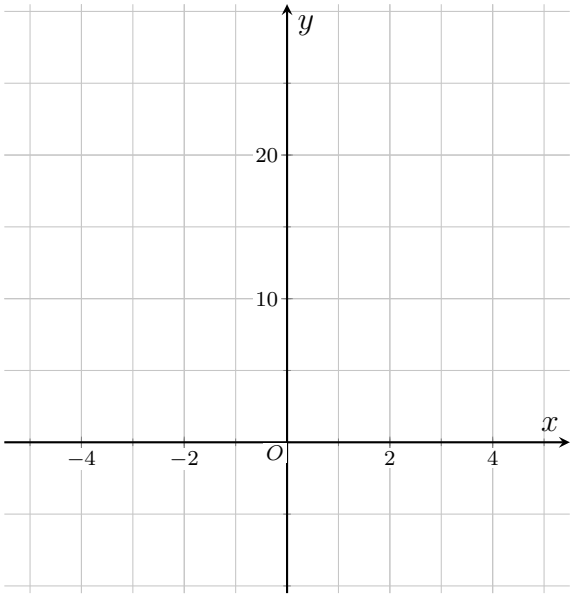
**Say it:** $f$ has a **relative minimum** near _____ and **relative maximums** near _____ and _____, so it has ____ **turning points**. Its **degree** is ____ and its **leading coefficient** is _____, so both ends _____.

## Try It 3: Graph a polynomial function

Consider the polynomial function $f(x) = x^5 + 18x^2 + 10x + 1$.

**a.** Make a table of values to identify key features and sketch a graph of the function.

| $x$ | $-3$ | $-2$ | $-1$ | $0$ | $1$ | $2$ | $3$ |
|---|---|---|---|---|---|---|---|
| $f(x)$ | | | | | | | |




Degree _____ leading coefficient _____

End behavior: as $x \to -\infty$, $y \to$ _____;

as $x \to \infty$, $y \to$ _____.

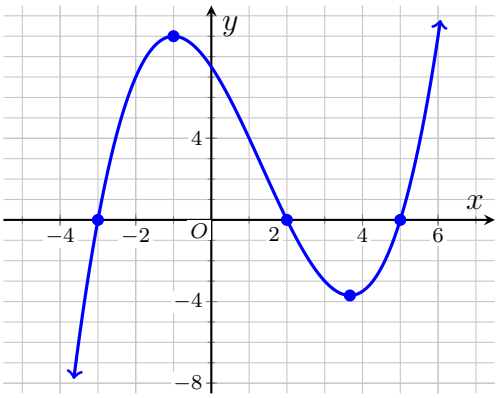
Turning points: between $x =$ _____ and $x =$ _____,

and between $x =$ _____ and $x =$ _____.

**b.** Find the average rate of change over the interval $[0, 2]$.

## Practice: Read a polynomial graph

The graph shows the polynomial function $f(x) = \frac{1}{4}(x+3)(x-2)(x-5)$. Its turning points are at $(-1, 9)$ and near $\left(\frac{11}{3}, -3.7\right)$.




Word bank: *decreasing*; *increasing*; *negative*; *positive*.

$f$ is _____ on the intervals $(-3, 2)$ and $(5, \infty)$.

$f$ is _____ on the interval $\left(-1, \frac{11}{3}\right)$.

$f$ is _____ on the intervals $(-\infty, -3)$ and $(2, 5)$.

$f$ is _____ on the intervals $(-\infty, -1)$ and $\left(\frac{11}{3}, \infty\right)$.

Which turning point is a **relative maximum**? How do you know?

## IXL: Practice for the quiz

Do the starred skills first. **SmartScore 70 or above on each starred (★) skill replaces your quiz score.**

| Code | Skill | Feeds |
|---|---|---|
| XMK ★ | Match polynomial functions and graphs using end behavior | Quiz 2: end behavior from the degree and the leading coefficient (Example 2); also choice B of Quiz 1 (full Savvas quiz only) |
| XJU ★ | Match polynomial functions and graphs | Quiz 3 and Quiz 4 (bonus): read a polynomial graph (Example 3) |
| DYB | Polynomial vocabulary | Warm-up: standard form, degree, leading coefficient |
| PHD | Average rate of change | Optional: Example 3B and Try It 3b (not on the quiz; Topic 3 test Question 9) |

## Exit question: Use end behavior to reason

An analyst for a new company used the first three years of revenue data to project future revenue for the company. The analyst predicts the function $f(x) = -2x^5 + 6x^4 - x^3 + 5x^2 + 6x + 50$ will give the revenue after $x$ years. Should the CEO expect the company to be successful? Explain.

The **degree** of $f$ is _____, which is _____, and the **leading coefficient** is _____, which is _____.

So the **end behavior** is: as $x \to \infty$, $y \to$ _____. In this context that means ...

_____

_____

_____

**Before the quiz, finish these sentences.**

The **degree** tells you whether the two ends of the graph point _____.

The sign of the **leading coefficient** tells you whether the right end goes _____.

A **turning point** is where the graph changes from _____ to _____, or the reverse.

A **relative maximum** is a turning point where the graph changes from _____ to _____.

*Want more?* Practice & Problem Solving 14 (DOK 3): write a polynomial in standard form with 6 terms, $y$-intercept at 1, and end behavior as $x \to -\infty$, $y \to +\infty$; as $x \to +\infty$, $y \to -\infty$. Explain how you chose it and graph it to check.